# Faddeev–Skyrme soliton energies from controlled lattice extrapolation

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## Abstract

We recompute the charge-one Faddeev–Skyrme soliton energy with both lattice systematics under explicit control and obtain $E_1/c_0 = 1.231 \pm 0.003$, an excess of 2.2% over the accepted 1.204 that we have been unable to remove. The systematics matter because they act in opposite directions—finite spacing lowers the energy, a finite periodic box raises it—so neither can be assessed alone; fitting $E = E_\infty - ah^2 + cL^{-p}$ to nine runs separates them. Four checks bear on whether the fault is ours. At matched spacing $h = 0.100$, the grid of the reference computation, we obtain 1.2257 with no extrapolation, a 1.8% gap of which our finite box can account for at most 0.7%. A fourth-order accurate symmetrisation, whose error has the opposite sign, extrapolates from above to 1.2371 against the second-order scheme's 1.2354 from below, the two agreeing to 0.14%. Adopting the reference's fixed boundaries in place of our periodic box shifts the continuum limit by 0.18%, and at finite volume widens the gap rather than closing it. And the identical procedure applied to the Skyrme model, where the charge-one energy follows from an ordinary differential equation, recovers that exact value to 0.03% from runs whose individual errors reach 2%. The discrepancy is confined to the absolute normalisation: we reproduce the published charge-one to charge-four configuration types, the scaling exponent 0.765 against 3/4, binding against fission in every sector, and, in the frame sector, binding energies per baryon within 0.15 percentage points of the tabulated values. We also record a discretisation pitfall with consequences beyond bookkeeping: any derivative whose symbol vanishes at the Nyquist frequency gives the discrete energy exact flat directions, along which the topological charge unwinds at no cost and the energy falls below the Vakulenko–Kapitanskii bound.

## I. INTRODUCTION

The Faddeev–Skyrme model [1] is the simplest field theory in three dimensions whose solitons are classified by a linking number rather than a winding number. For a unit three-vector field $\boldsymbol{n} : \mathbb{R}^3 \to S^2$ with $\boldsymbol{n} \to \boldsymbol{n}_0$ at infinity, configurations fall into sectors labelled by the Hopf invariant $Q_H \in \pi_3(S^2) = \mathbb{Z}$, and the minimal energy in each sector is a well-posed numerical question that has been studied for two decades [2–7]. The charge-one energy is the reference against which all other sectors are quoted, and it is usually taken to be 1.204 in the normalisation fixed by Ward's conjectured bound.

This paper recomputes that number with the two lattice systematics separated and extrapolated, and reports that we cannot reproduce it. Our value is 2.2% higher. Because a discrepancy of this size is more often a defect of the computation than of the literature, most of what follows is an attempt to break our own result.

Two things make the exercise worth reporting. The first is a discretisation pitfall that is easy to fall into and whose symptom is dramatic—the topological charge simply dissolves—but which we have not seen stated in these terms. The second is that the two lattice systematics push in opposite directions, so a study that varies only the spacing, or only the box, will appear better converged than it is. Section III treats the first, Section IV the second.

Section V is the control. Applied to the Skyrme model, where the charge-one energy is available from an ordinary differential equation, the same code and the same extrapolation recover the exact answer to three parts in ten thousand. Section VI reports the $Q_H \leq 4$ spectrum, which agrees with the published one in every structural respect. All source is available [8].

## II. MODEL, BOUND AND NORMALISATION

The static energy is

$$E[\boldsymbol{n}] = \int d^3x \left[ c_2\, \partial_i \boldsymbol{n} \cdot \partial_i \boldsymbol{n} + \tfrac{c_4}{2} \left(\partial_i \boldsymbol{n} \times \partial_j \boldsymbol{n}\right) \cdot \left(\partial_i \boldsymbol{n} \times \partial_j \boldsymbol{n}\right) \right], \tag{1}$$

and we set $c_2 = c_4 = 1$ throughout, so that lengths and energies are in model units. Under $\boldsymbol{n}(x) \to \boldsymbol{n}(x/\mu)$ the two terms scale as $E(\mu) = \mu E_2 + \mu^{-1} E_4$, so every stationary point satisfies the virial identity $E_2 = E_4$; with either constant absent the soliton collapses or spreads, and Derrick's theorem is satisfied rather than evaded.

Finite-energy configurations obey the bound of Vakulenko and Kapitanskii [9],

$$E \geq c_0 |Q_H|^{3/4}, \tag{2}$$

whose proof does not fix the constant. Ward [6] conjectured the sharp value, which in the normalisation of (1) reads

$$c_0 = 32\pi^2\sqrt{2} = 446.65. \tag{3}$$

Reference [7] divides its energies by exactly this factor and states the conjecture with unit coefficient; its integrand is identical to (1), so its tabulated energies are directly comparable with ours with no conversion. We have verified the identity $\frac{1}{2}(\partial_i \boldsymbol{n} \times \partial_j \boldsymbol{n})^2 = \frac{1}{2}[(\partial_i \boldsymbol{n} \cdot \partial_i \boldsymbol{n})^2 - (\partial_i \boldsymbol{n} \cdot \partial_j \boldsymbol{n})^2]$ symbolically, so the two normalisations agree term by term.

Because the exponent 3/4 is sub-linear, $E_Q < |Q|E_1$ is permitted and multi-charge solitons may be bound against fission; whether they are is settled in Sec. VI. For the general theory we refer to Manton and Sutcliffe [10].

## III. LATTICE CONSTRUCTION, AND A NULL MODE THAT DESTROYS TOPOLOGY

We minimise (1) on a periodic cubic lattice. One implementation choice carries more than numerical significance.

With central differences *of any order*, the Fourier symbol of the derivative operator vanishes at the Nyquist wavenumber. For the fourth-order antisymmetric stencil, setting $f_{i\pm1} \to -f_i$ and $f_{i\pm2} \to f_i$ gives identically zero. A checkerboard mode therefore costs *no energy*. The discrete energy acquires exact flat directions, and gradient flow carries any Hopfion monotonically to the vacuum: in our first runs $Q_H$ fell from 1 to 0 while $E$ fell far below the bound $c_0|Q_H|^{3/4}$, which is supposed to be inviolable. The topological protection is lost not because the physics permits it but because the discretisation has a zero-cost path out of the sector.

The cure is a scheme whose symbol vanishes only at $k = 0$. We use one-sided differences and symmetrise the *energy* rather than the derivative,

$$E = \tfrac{1}{2}\big[E(D^+\boldsymbol{n}) + E(D^-\boldsymbol{n})\big], \qquad (D^\pm f)_i = \pm\frac{f_{i\pm1} - f_i}{h}, \tag{4}$$

which has no null mode and, because $(D^\pm)^\mathsf{T} = -D^\mp$ holds exactly on a periodic lattice, makes the analytic variation the *exact* gradient of the discrete energy. We verify this against finite differences of the energy density to a maximum relative error of $5 \times 10^{-7}$.

The symmetrisation also fixes the form of the extrapolation used below. Expanding $D_i^\pm = \partial_i \pm \frac{h}{2}\partial_i^2 + \frac{h^2}{6}\partial_i^3 + \ldots$ and averaging removes the $O(h)$ term, leaving

$$\tfrac{1}{2}\int \big[|D^+\boldsymbol{n}|^2 + |D^-\boldsymbol{n}|^2\big] = \int |\partial\boldsymbol{n}|^2 - \frac{h^2}{12}\int |\partial^2\boldsymbol{n}|^2 + O(h^4), \tag{5}$$

after integrating $\int \partial\boldsymbol{n} \cdot \partial^3\boldsymbol{n} = -\int |\partial^2\boldsymbol{n}|^2$ by parts. The discrete energy is therefore below the continuum one, by $O(h^2)$ – both the sign and the order seen in the fits of Sec. IV.

TABLE I. Lattice-spacing dependence of the $Q_H = 1$ soliton at $L = 12$.

| $N$ | $h$ | $E$ | $E/c_0$ | $Q_H$ | $E_2/E_4$ |
|---|---|---|---|---|---|
| 56 | 0.2143 | 532.28 | 1.1917 | 0.9833 | 0.823 |
| 72 | 0.1667 | 540.42 | 1.2100 | 0.9951 | 0.883 |
| 88 | 0.1364 | 544.12 | 1.2182 | 0.9980 | 0.904 |
| 96 | 0.1250 | 545.29 | 1.2208 | 0.9986 | 0.908 |
| 104 | 0.1154 | 546.18 | 1.2228 | 0.9990 | 0.910 |
| 120 | 0.1000 | 547.45 | 1.2257 | 0.9994 | 0.911 |

*A fourth-order variant.* Nothing forces the one-sided differences in (4) to be first-order accurate. Built from third-order one-sided differences,

$$(D_3^+ f)_i = \frac{-11f_i + 18f_{i+1} - 9f_{i+2} + 2f_{i+3}}{6h}, \tag{6}$$

and its reflection, the leading errors are equal and opposite, $D_3^\pm = \partial \pm \frac{h^3}{4}\partial^4 + \ldots$, so they cancel in the symmetrised energy and leave an $O(h^4)$ scheme. Its Nyquist symbol is $-20/3h$ rather than zero, so the null mode is still absent, and $(D_3^+)^\mathsf{T} = -D_3^-$ still holds, so the variation remains exact. We refer to the two schemes as $O(h^2)$ and $O(h^4)$; crucially their errors have opposite sign, so together they bracket the continuum limit.

The Hopf charge is evaluated by reconstructing $\boldsymbol{A}$ spectrally in Coulomb gauge from $\boldsymbol{B}$, with $\|\nabla \times \boldsymbol{A} - \boldsymbol{B}\|/\|\boldsymbol{B}\| < 10^{-4}$. Seeds of known charge come from composing a degree-$d$ hedgehog with the Hopf projection (giving $Q_H = d$, since the Chern–Simons three-form pulls back with the degree) and from the toroidal $A_{n,m}$ ansatz [3, 5] (giving $Q_H = nm$).

## IV. THE CHARGE-ONE ENERGY

Two systematics act in opposite directions, and both must be removed before any comparison is meaningful.

*Finite spacing lowers the energy*, by (5). Repeating the $Q_H = 1$ relaxation at fixed box $L = 12$, with every lattice initialised by resampling the converged solution so that all points share the same convergence quality, gives Table I. The measured charge approaches the integer and the virial ratio approaches unity, both as they must.

*Finite volume raises it.* The director is massless, so the soliton has power-law tails and interacts with its own periodic images. Holding $h = 1/6$ and enlarging the box gives $E = 540.50$, $538.77$, $538.40$ for $L = 12, 16, 20$, with the energy fraction within six cells of the boundary falling from $4.0 \times 10^{-3}$ to $1.3 \times 10^{-4}$.

Neither series alone is a convergence test, because the two effects partially cancel. Fitting them together,

$$\frac{E}{c_0} = E_\infty - a\,h^2 + c\,L^{-p}, \tag{7}$$

to all nine runs gives an $h^2$ coefficient $a = 0.94$, a box correction of $+0.4$ to $+0.7\%$ at $L = 12$, and

$$E_1 = 550 \pm 2, \qquad E_1/c_0 = 1.231 \pm 0.003, \tag{8}$$

with residuals of $4 \times 10^{-4}$ throughout. The quoted spread is the variation of the intercept as the box exponent is fixed anywhere between $p = 2$ and $p = 6$, widened to cover a second fit to an independently relaxed set of runs. The extrapolation reaches only 0.4% beyond the finest lattice computed.

Against the published 1.204 this is an excess of 2.2%. Three checks bear on whether the fault is ours.

*Matched resolution.* Our finest lattice uses $h = 0.100$, exactly the spacing of Ref. [7], and returns 1.2257 with no extrapolation at all. At equal spacing the gap is 1.8%, of which our finite box can account for at most 0.7%.

*An independent discretisation.* The remaining worry is that our energy is only $O(h^2)$ accurate while Ref. [7] uses fourth-order derivatives, so that the extrapolation rather than the data is at fault. Repeating the whole calculation with the $O(h^4)$ scheme of Sec. III, whose error enters with the opposite sign, gives Table II: the two schemes approach from opposite sides and agree with each other to 0.14%. Two discretisations of different order converging from opposite directions do not conspire.

*A control with a known answer.* This is Sec. V.

## V. A CONTROL: THE FRAME SECTOR

The Faddeev–Skyrme model has no closed-form solution, so within it a lattice can only be compared against other lattices. The Skyrme model does have one, and it is close enough

TABLE II. The two symmetrised schemes at $L = 12$. The $O(h^2)$ scheme approaches from below and the $O(h^4)$ scheme from above; the extrapolations agree to 0.14%. Values are $E/c_0$ before the box correction of Eq. (7).

| $h$ | $O(h^2)$ | $E_2/E_4$ | $O(h^4)$ | $E_2/E_4$ |
|---|---|---|---|---|
| 0.1875 | 1.2026 | 0.862 | 1.2646 | 1.071 |
| 0.1500 | 1.2147 | 0.902 | 1.2492 | 0.975 |
| 0.1250 | 1.2207 | 0.920 | 1.2420 | 0.943 |
| $\to 0$ | **1.2354** | | **1.2371** | |

to be a genuine control: the same energy functional, the same code, the same boundary conditions and the same extrapolation, with only the target changed from $S^2$ to $S^3$.

For $\varphi : \mathbb{R}^3 \to S^3$ the energy (1) is the Skyrme model [11] in Manton's strain form [12], configurations are classified by $\pi_3(S^3) = \mathbb{Z}$ with baryon number

$$B = \frac{1}{2\pi^2} \int \det[\varphi, \partial_1\varphi, \partial_2\varphi, \partial_3\varphi]\, d^3x, \tag{9}$$

and the topological bound follows from two applications of AM–GM to the principal stretches, $E \geq 12\pi^2\sqrt{c_2 c_4}\,|B|$. For $B = 1$ the minimiser is spherically symmetric, so with $\varphi = (\cos f, \sin f\, \hat{\boldsymbol{r}})$ the energy reduces to

$$E = 4\pi \int_0^\infty \left[ f'^2 r^2 + 2\sin^2\! f\,(1 + f'^2) + \frac{\sin^4\! f}{r^2} \right] dr, \tag{10}$$

whose Euler–Lagrange equation we solve as a two-point boundary value problem with $f(0) = \pi$, $f(\infty) = 0$, obtaining

$$E/12\pi^2 = 1.23145, \tag{11}$$

consistent with the value 1.232 standard since Ref. [13]. Because we compute (11) ourselves it tests the lattice rather than our reading of a table.

Relaxing on the lattice at two box sizes and three spacings each, and fitting Eq. (7) to all six runs, gives

$$E_\infty/12\pi^2 = 1.2311, \tag{12}$$

with an rms residual of $6.5 \times 10^{-5}$ and a leave-one-out spread of $\pm 0.0008$. This is 0.03% from the exact answer (11), obtained from runs whose individual errors reach 2%. The box

TABLE III. Frame-sector solitons at $L = 12$, $N = 96$, from rational-map seeds of the indicated symmetry. Literature values are Ref. [14]. Binding is $1 - E_B/(BE_1)$.

| $B$ | symmetry | $E/12\pi^2 B$ | Ref. [14] | binding | lit. binding |
|---|---|---|---|---|---|
| 1 | spherical | 1.2231 | 1.2322 | — | — |
| 2 | axial | 1.1712 | 1.1791 | 4.25% | 4.31% |
| 3 | tetrahedral | 1.1396 | 1.1462 | 6.83% | 6.98% |
| 4 | cubic | 1.1110 | 1.1201 | 9.17% | 9.10% |

exponent comes out $p \simeq 2.2$, and the $L = 8$ series overshoots the exact value while still rising with resolution—direct confirmation that a periodic box biases upward, an effect also noted in the Skyrme literature [14].

A procedure accurate to three parts in ten thousand on a problem with a known answer is unlikely to be wrong by 2% on a neighbouring one. This is our main reason for reporting the discrepancy of Sec. IV as a discrepancy.

The same machinery reproduces the multi-baryon sector. Table III gives $B = 1$–4 from symmetric rational-map seeds [15]; the energies sit 0.6–0.8% below the tabulated values [14], uniformly, so the binding energies—being ratios, in which the systematic cancels—come out within 0.15 percentage points.

## VI. THE CHARGE-ONE TO CHARGE-FOUR SPECTRUM

Table IV gives the Hopf spectrum on a common lattice, and Table V compares it with Ref. [7]; Fig. 1 shows the relaxed configurations and Fig. 2 the energies against the bound and the fission threshold.

Four structural results, none of them imposed.

*Every multi-charge sector is bound against fission*, $E_Q < QE_1$ by 18.5%, 23.4% and 28.0% for $Q = 2, 3, 4$.

*The sub-linear exponent is recovered.* A power-law fit gives $E_Q \propto |Q_H|^{0.765}$ against the Vakulenko–Kapitanskii 3/4; the same fit to the published energies gives 0.760.

*The $Q_H = 4$ ground state is linked.* The $A_{2,2}$ seed relaxes to 1563.2, below both the hedgehog $d$=4 (1621.6) and $A_{4,1}$ (1621.8), and it is more compact than the $Q_H = 3$ soliton

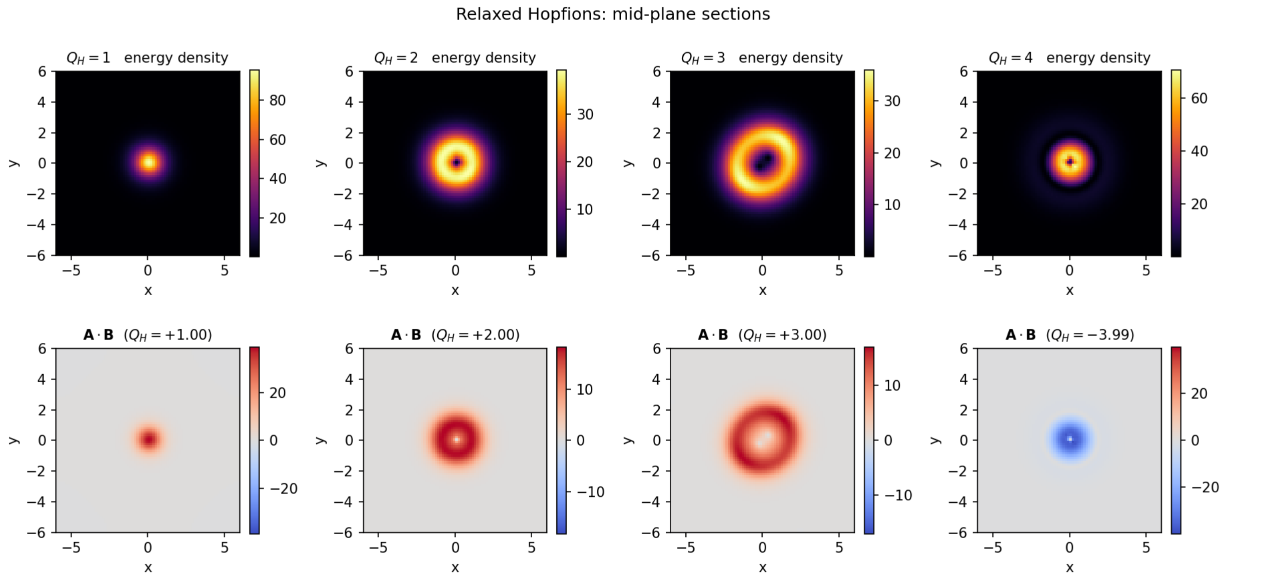


FIG. 1. Mid-plane sections of the relaxed Hopfions. Top: energy density. Bottom: Hopf density $\boldsymbol{A} \cdot \boldsymbol{B}$ on a symmetric colour scale, so the vacuum is neutral in every panel; the $Q_H = 4$ solution was found with negative charge, which is equivalent to the positive one by a spatial reflection. The $Q_H = 2, 3$ ground states are toroidal; the $Q_H = 4$ ground state is the more compact linked $A_{2,2}$ configuration.

TABLE IV. Relaxed Hopfion spectrum ($N = 80$, $L = 12$, $h = 0.15$). Several independent seeds were relaxed in each sector and the lowest is quoted.

| $Q_H$ | seed | $E$ | $E/c_0\|Q\|^{3/4}$ | $R_{\rm rms}$ | $E_2/E_4$ | $E_Q/(QE_1)$ |
|---|---|---|---|---|---|---|
| 1 | hedgehog $d$=1 | 542.65 | 1.215 | 1.705 | 0.887 | 1.000 |
| 2 | hedgehog $d$=2 | 884.15 | 1.177 | 2.089 | 0.950 | 0.815 |
| 3 | hedgehog $d$=3 | 1247.30 | 1.225 | 2.489 | 0.955 | 0.766 |
| 4 | $A_{2,2}$ (linked) | 1563.24 | 1.237 | 2.331 | 0.956 | 0.720 |

($R_{\rm rms} = 2.33$ against 2.49). Linking, not size, lowers the energy.

*Seeds that should agree do.* $A_{2,1} \to 884.20$ against the hedgehog's 884.15, and $A_{3,1} \to 1248.34$ against 1247.30: agreement at the $10^{-3}$ level from entirely different initial data. The $A_{1,m}$ family stalls higher (1006.9, 1478.9, 1898.9), which we report as upper stationary configurations without claiming they are local minima.

The excess over Ward's bound is 21.5, 17.7, 22.5 and 23.7% here against 20.4, 17.0, 20.8

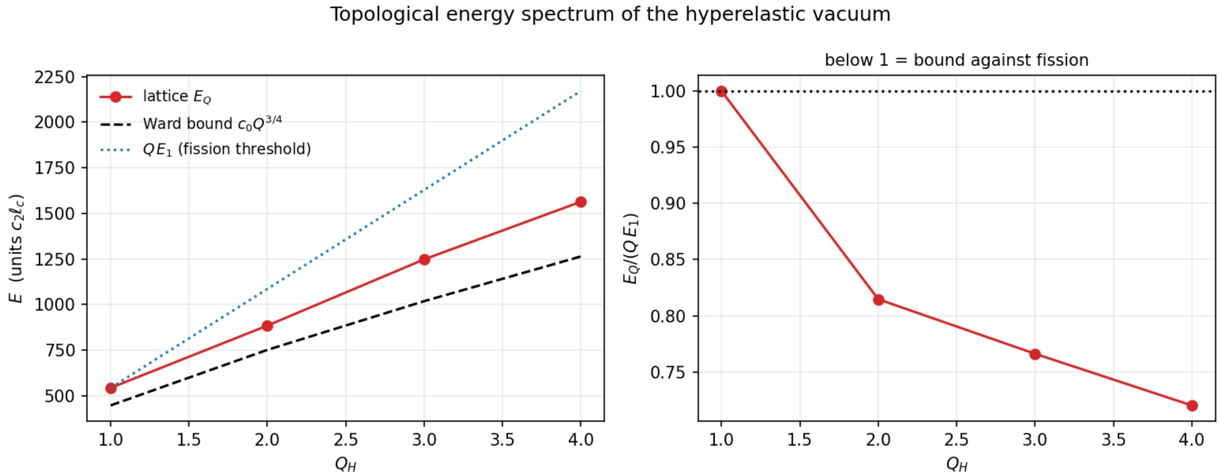


FIG. 2. Left: computed energies against the Ward bound $c_0Q^{3/4}$ and the fission threshold $QE_1$. Right: $E_Q/(QE_1)$; values below unity indicate binding against fission.

TABLE V. Comparison with Ref. [7] in the common normalisation $E/c_0$, on the lattice of Table IV. The ground-state configuration type is reproduced in every sector.

| $Q_H$ | this work | Ref. [7] | difference | type (ours / published) |
|---|---|---|---|---|
| 1 | 1.215 | 1.204 | +0.9% | $A_{1,1}$ / $A_{1,1}$ |
| 2 | 1.980 | 1.967 | +0.6% | $A_{2,1}$ / $A_{2,1}$ |
| 3 | 2.793 | 2.754 | +1.4% | $A_{3,1}$ / $\widetilde{A}_{3,1}$ |
| 4 | 3.500 | 3.445 | +1.6% | $A_{2,2}$ / $A_{2,2}$ |

and 21.8% in Ref. [7]: not only the overall $\sim 20\%$ level but the characteristic dip at $Q_H = 2$ is reproduced. The structure of the spectrum is therefore not in question; the disagreement is confined to the absolute normalisation.

## VII. DISCUSSION

We are unable to reduce the charge-one discrepancy below 2%, and we have not found a defect in our own calculation that would produce it. The candidate explanations we can test have been tested: the spacing, by matching Ref. [7] exactly; the order of the scheme, by an independent $O(h^4)$ discretisation approaching from the other side; the box, by an explicit two-parameter fit; the boundary condition, by repeating the calculation with Dirichlet walls;

and the whole procedure, by a control with an exact answer.

The remaining methodological difference is the boundary condition itself. Reference [7] truncates the field to the vacuum at the boundary of a box of side 15.1; we use a periodic box. For a massless field with power-law tails the natural expectation is that truncation removes gradient energy a periodic box retains, with the sign right to explain part of a gap. We have measured it rather than argued about it.

Repeating the calculation under Dirichlet walls—stencils stopping at the boundary rather than wrapping, with a collar four cells deep held at the vacuum, wider than the deepest stencil—gives the paired comparison of Table VI. Within a row the seed, the grid and the relaxation schedule are identical, so the only difference is the boundary. The expectation is wrong in sign: fixed walls *raise* the energy at every box size, because clamping the tail forbids it from relaxing, which costs more than truncation saves. The excess falls monotonically with box size, from 0.69% at $L = 10$ to 0.135% at $L = 14$, as two conventions approaching a common limit require. Fitting (7) separately to each convention gives

$$E_\infty^{\rm per} = 1.2280 \pm 0.0008, \qquad E_\infty^{\rm fix} = 1.2257 \pm 0.0012, \tag{13}$$

agreeing to $1.6\sigma$. The boundary condition moves the continuum limit by 0.18%—an order of magnitude too little to account for the discrepancy. Under the reference's own boundary condition the excess over 1.204 is 1.81%. The periodic member of this series also reproduces (8) within its uncertainty, on an independent ladder of grids, which is a consistency check on both.

Against all of this, Ref. [7] itself argues the opposite way, noting that its grid "appear[s] to underestimate the energy by an amount of the order of 1%" and quoting $E \simeq 1.21$ from higher-resolution studies. Our result would put the underestimate nearer 2%.

We therefore report $E_1/c_0 = 1.231 \pm 0.003$ as our result and the difference from 1.204 as open, with the boundary condition no longer among the candidate explanations. What would settle it is an independent implementation rather than an independent run of ours; the code, including both schemes and both boundary conventions, is available [8].

TABLE VI. Paired boundary comparison at $Q_H = 1$. Each row is two relaxations from the same seed on the same grid with the same schedule, differing only in the boundary condition. Fixed walls lie above periodic everywhere, and the gap closes as the box grows.

| $L$ | $h$ | periodic | fixed | difference (%) |
|---|---|---|---|---|
| 10 | 0.1471 | 1.2209 | 1.2294 | +0.693 |
| 10 | 0.1250 | 1.2264 | 1.2341 | +0.629 |
| 10 | 0.1000 | 1.2316 | 1.2382 | +0.533 |
| 12 | 0.1500 | 1.2148 | 1.2193 | +0.372 |
| 12 | 0.1250 | 1.2210 | 1.2250 | +0.329 |
| 12 | 0.1000 | 1.2264 | 1.2293 | +0.242 |
| 14 | 0.1489 | 1.2124 | 1.2149 | +0.209 |
| 14 | 0.1250 | 1.2182 | 1.2204 | +0.185 |
| 14 | 0.1094 | 1.2218 | 1.2235 | +0.135 |


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Numerical work used NumPy, SciPy, SymPy, Matplotlib and CuPy. The author used an AI assistant (GitHub Copilot) for code development, numerical analysis and manuscript preparation; all claims, and responsibility for them, are the author's.


## Appendix A: Numerical methods

*Discretisation.* Periodic cubic lattice; the energy is the symmetrisation of Eq. (4) in either the $O(h^2)$ or the $O(h^4)$ form. Because $(D^{\pm})^{\mathsf{T}} = -D^{\mp}$, the analytic variation is the exact gradient of the discrete energy, verified against finite differences of the energy density to $5 \times 10^{-7}$.

*Minimisation.* Adaptive damped flow: the step is capped so that no site rotates by more than $\delta$ radians per iteration; a heavy-ball term with energy arrest accelerates convergence; and the momentum is projected back onto the tangent space each step—omitting this projection allows a spurious radial component to accumulate and drives the field out of its sector. Every hundred steps the exact Derrick dilation $\mu = \sqrt{E_4/E_2}$ is applied and accepted only if it lowers the energy *and* does not push the configuration into the boundary collar; the second condition matters, because unbounded dilation samples only the core and returns a uniform field, whose energy is zero and which the energy test alone would therefore accept.

*Convergence.* Stationarity is checked by continued relaxation: the stored $Q_H = 1$ field moves by $2 \times 10^{-4}$ in relative energy over 1800 further steps, and the residual virial deficit is a property of the discrete functional rather than of incomplete relaxation, since it persists while the energy is stationary and falls with both $h$ and $L$.

*Topological charge.* $\boldsymbol{B}$ from fourth-order central differences, $\boldsymbol{A}$ from $\hat{\boldsymbol{A}} = i\boldsymbol{k} \times \hat{\boldsymbol{B}}/k^2$ in Coulomb gauge; the residual $\|\nabla \times \boldsymbol{A} - \boldsymbol{B}\|/\|\boldsymbol{B}\|$ is below $10^{-4}$ at $N = 96$. The baryon number of the frame sector is the determinant of $[\varphi, \partial_1\varphi, \partial_2\varphi, \partial_3\varphi]$ evaluated at every site, and is integral to $5 \times 10^{-4}$ at the finest lattice.

### Appendix B: Reproducibility

The supplement [8] comprises a library and one driver per result, with `verify_claims.py` re-deriving every quoted quantity from the stored fields rather than from intermediate summaries, and `verify_manuscript.py` checking that the numbers printed in this paper are the numbers those drivers produced. The lattice kernels are written against a swappable array backend and run unchanged on a CPU or, with `--gpu`, on a CUDA device; double precision is used throughout, since the kernels are limited by memory bandwidth rather than by floating-point throughput. The GPU path agrees with the CPU path to rounding for every energy, charge and radius reported here, except in the Derrick rescaling step, where the host and device cubic-spline interpolators differ at the $10^{-7}$ level; because that step is accepted only when it lowers the energy, this perturbs the relaxation path but not the minimum reached. All numbers quoted in this paper were produced on the CPU path, with the single exception of the paired boundary comparison of Table VI, which was run on the GPU. That comparison is between two relaxations sharing a code path, so the interpolator difference enters both members of a pair alike and cancels in the quantity of interest.

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